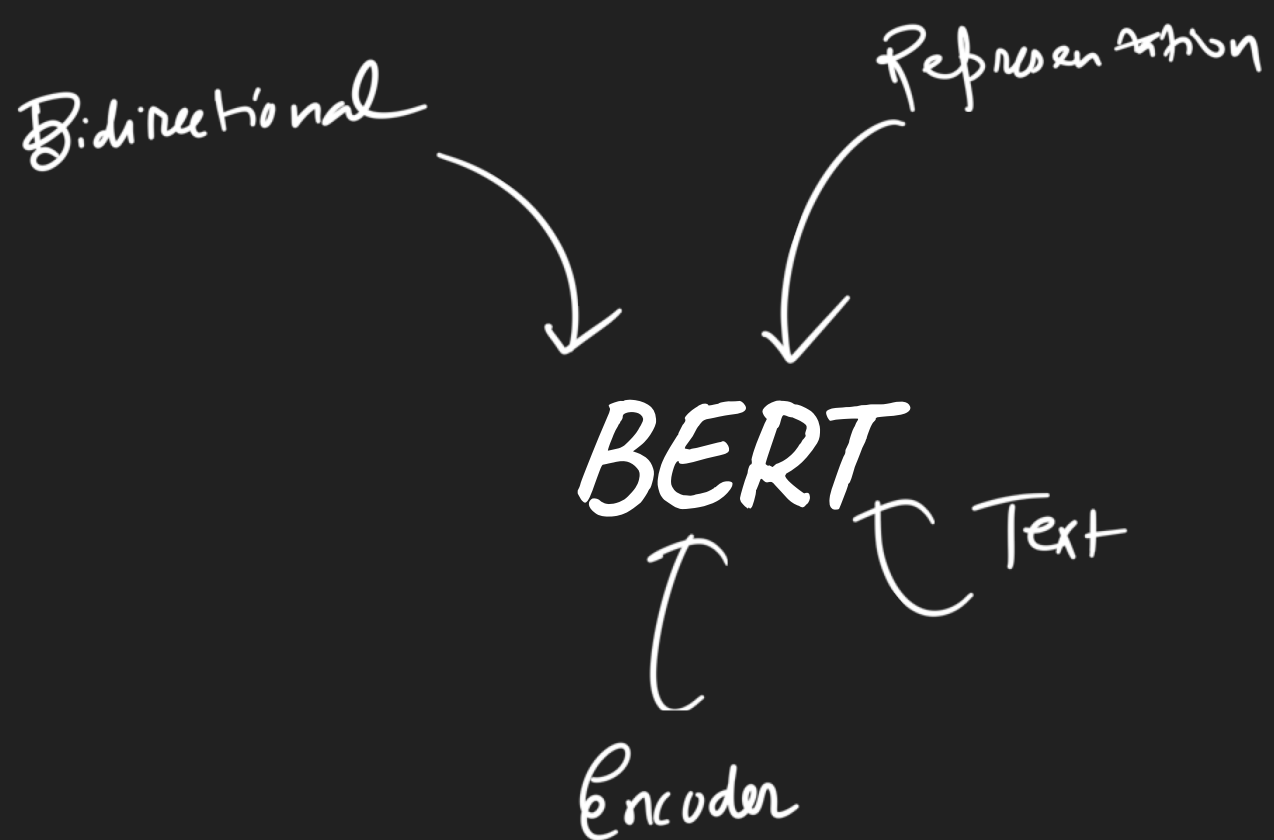
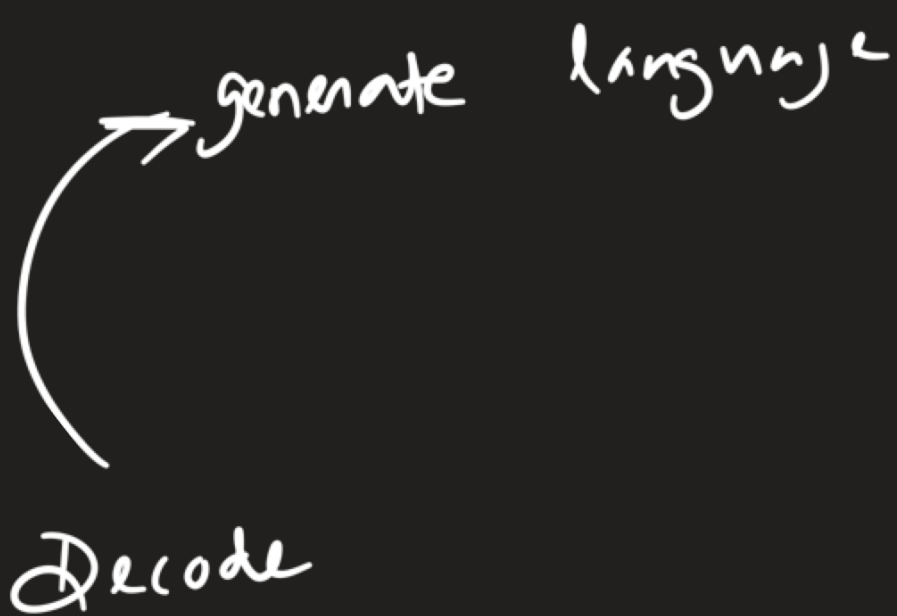






Encoder



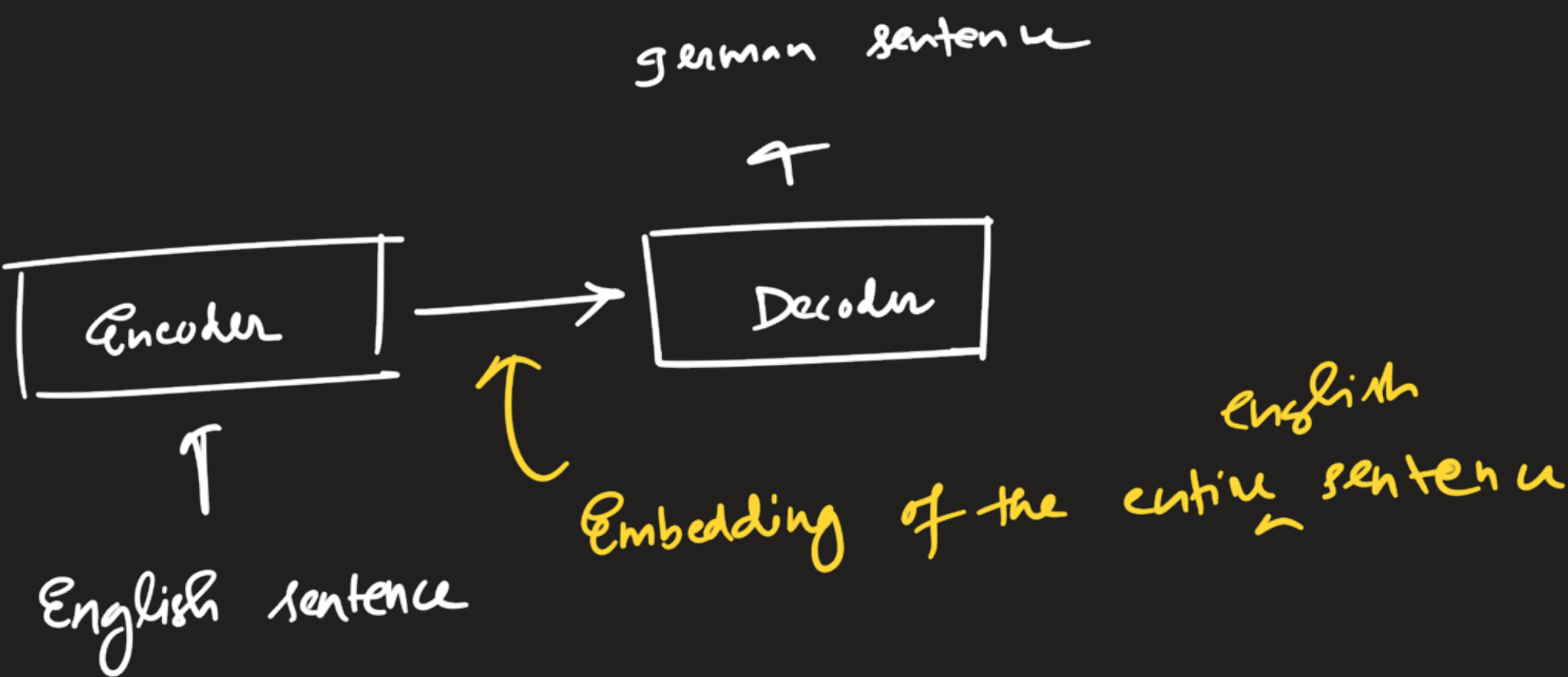
Decode

generate language

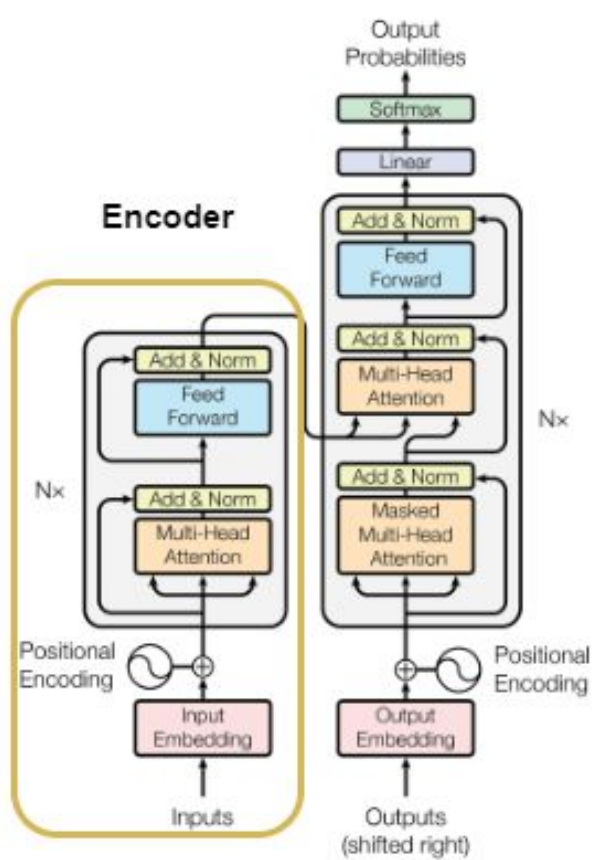
Wp Roses are red, violets are blue



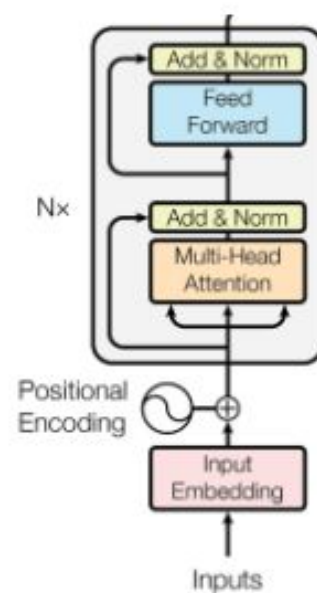
~~This song is good~~
~~This song is good~~



Transformers



Transformer Encoder



Bert-Base

Bert-Large

Encoder 1

Encoder 1

Encoder 2

Encoder 2

Encoder 11

Encoder 11

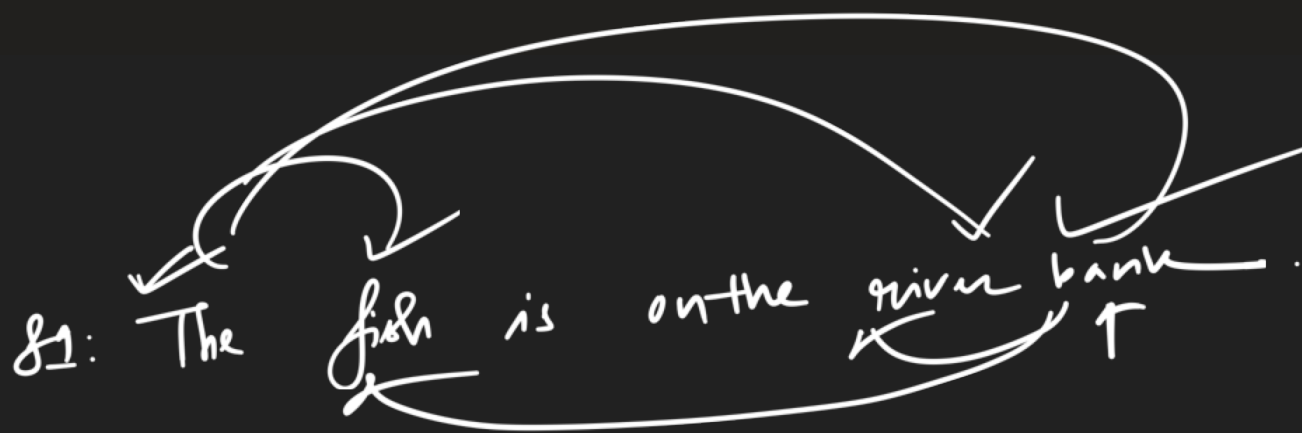
Encoder 12

Encoder 12

Encoder 23

Encoder 24

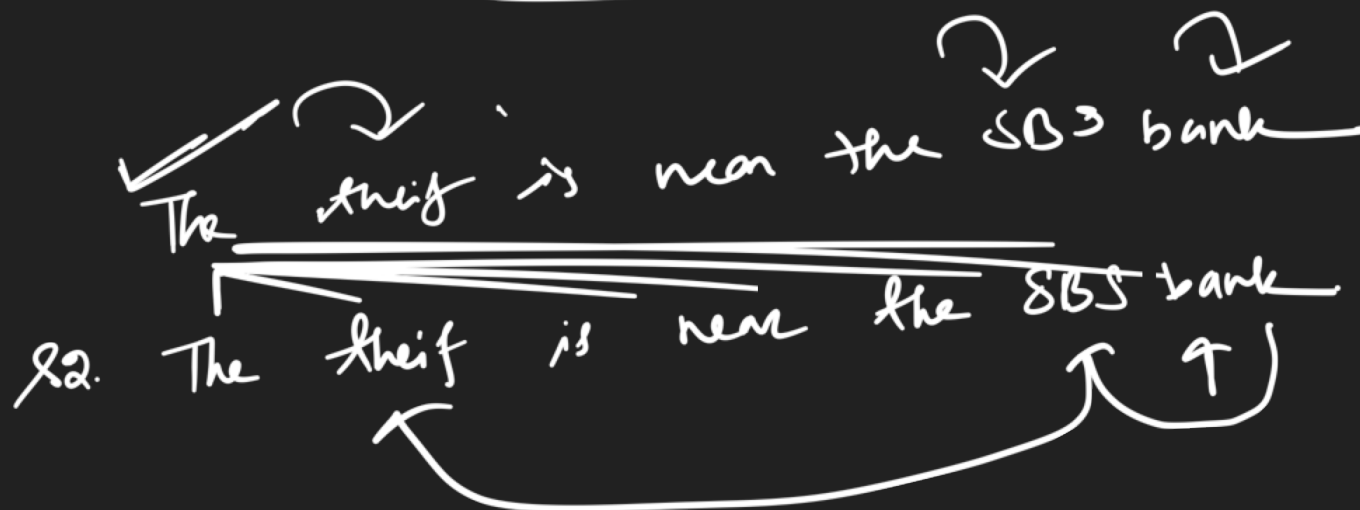
Q1: The fish is on the river bank.



WordVec

The thief is near the SBS bank

Q2. The thief is near the SBS bank



Embeddings which are contextualized

Encoder rep

~~He~~

He is playing → he is play ##ing

Corpus of text

✓ play
beg
beginning
playing

4

8

→ play ✓
→ beg ✓
→ beg #ing
→ play #ing
playing

3 tokens

playing ← frequently occurring subword
hugging
loosening
begging

— play #ing

drain
rain
train

✓ # ain

✓ a

✓ in

train = / #n #a #in

Aman is → play #ing
↑
playing

g	→	19
am	→	200
good	→	300

{

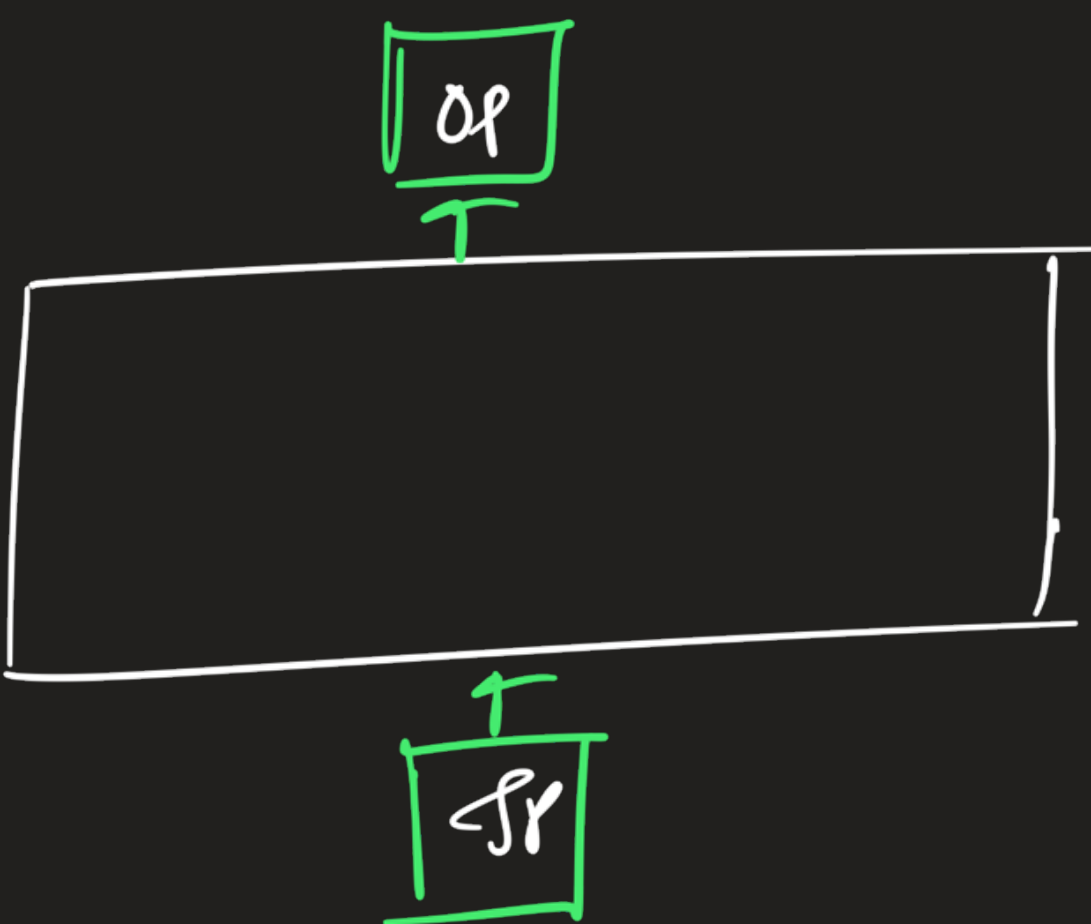
19: g,
200: am,
300: good

}

playing —
luggin —

play has ing

token →  → vector



Agenda

ip of BERT

op

Pre-training Loss

Fine-tune Loss

Tasks

Word2Vec → WordPiece

S1 - my dog is cute ..
S2 - he likes playing ..

input

✓ CLS (special token)

full seq

WordPiece

Position Embeds

Input	[CLS]	my	dog	is	cute	[SEP]	he	likes	play	##ing	[SEP]
Token Embeddings	$E_{[CLS]}$	E_{my}	E_{dog}	E_{is}	E_{cute}	$E_{[SEP]}$	E_{he}	E_{likes}	E_{play}	$E_{\#ing}$	$E_{[SEP]}$
	+	+	+	+	+	+	+	+	+	+	+
Segment Embeddings	E_A	E_A	E_A	E_A	E_A	E_A	E_B	E_B	E_B	E_B	E_B
	+	+	+	+	+	+	+	+	+	+	+
Position Embeddings	E_0	E_1	E_2	E_3	E_4	E_5	E_6	E_7	E_8	E_9	E_{10}

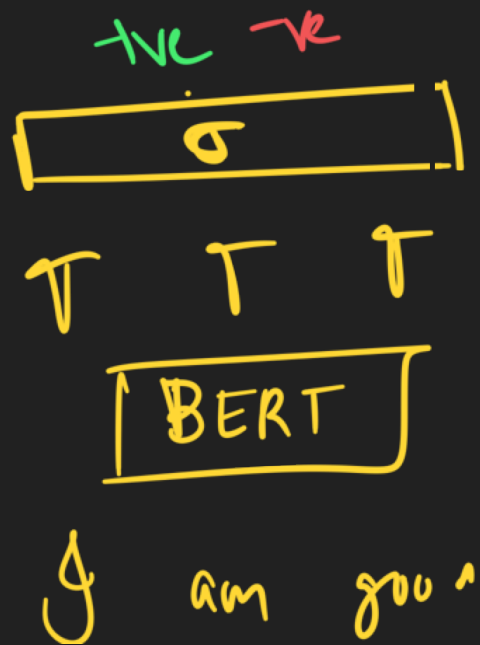
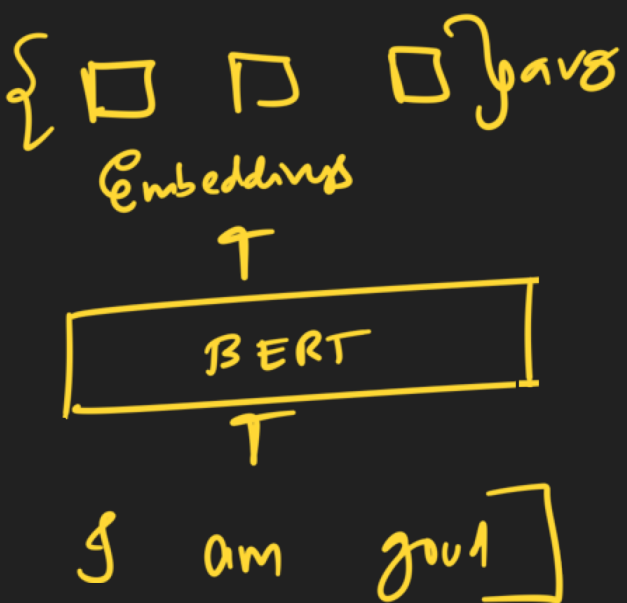
} demarcates two sentences

E_{CLS} E_{my}

Two special tokens - CLS
SEP

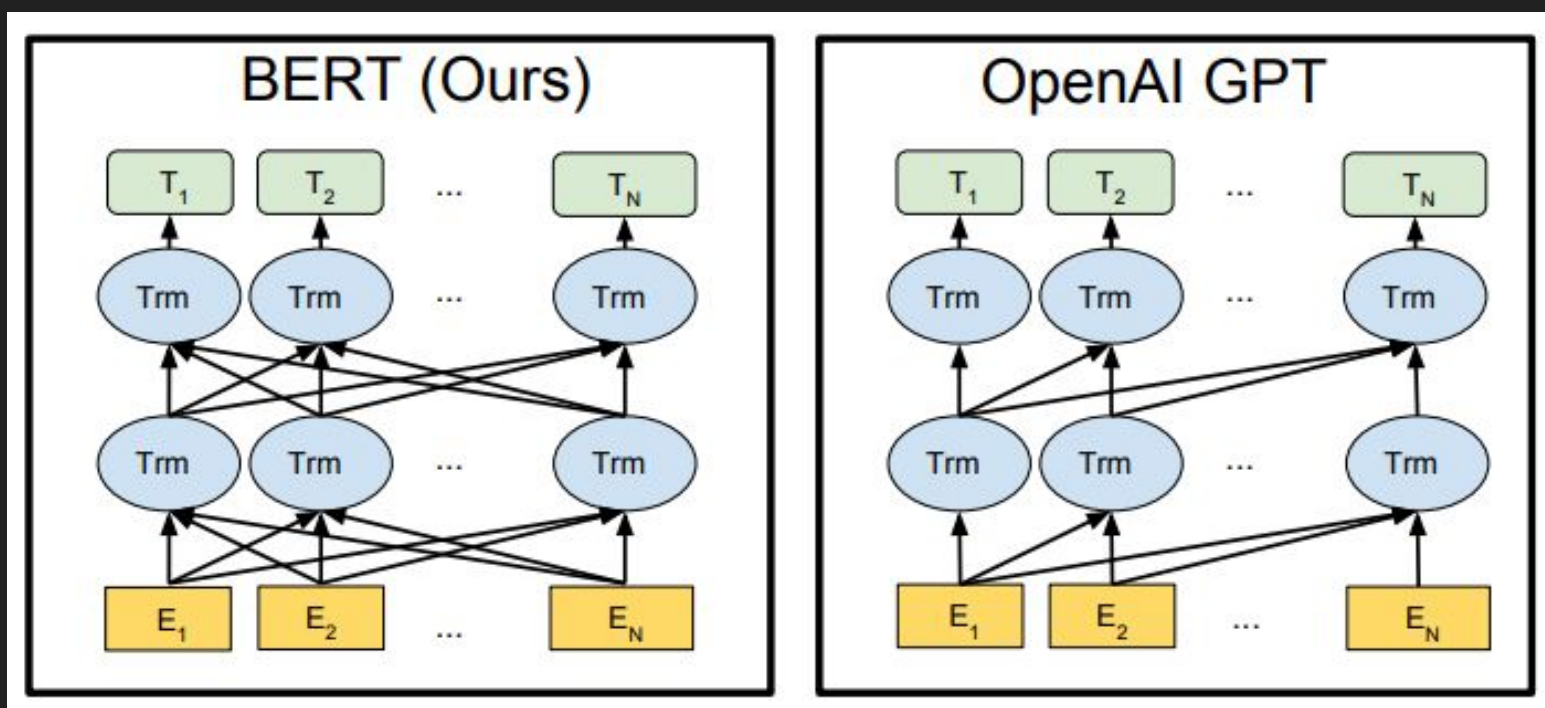
fine tuning

BERT -> A Word Embedding or Pretrained Model



Bidirectional

Unidirectional



This song is good

BERT Training

acquired data / compute

BERT is pretrained by Google

Big tech

Step 1

Pretraining

- make the model understand a language or a domain
- understand a language or a domain
- lots of data (open source)
- || || compute

Finetune

~~downstream task~~

Data is customized

sentiment classification

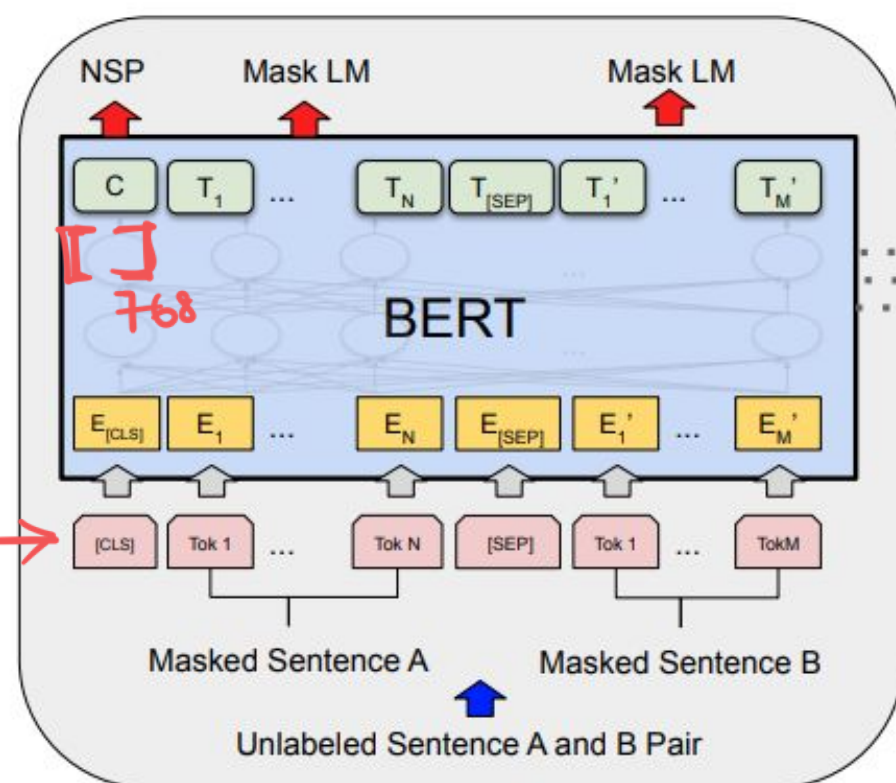
Step 2

Input/Output Representations To make BERT handle a variety of down-stream tasks, our input representation is able to unambiguously represent both a single sentence and a pair of sentences (e.g., $\langle \text{Question}, \text{Answer} \rangle$) in one token sequence. Throughout this work, a “sentence” can be an arbitrary span of contiguous text, rather than an actual linguistic sentence. A “sequence” refers to the input token sequence to BERT, which may be a single sentence or two sentences packed together.

We use WordPiece embeddings (Wu et al., 2016) with a 30,000 token vocabulary. The first token of every sequence is always a special classification token ($[\text{CLS}]$). The final hidden state corresponding to this token is used as the aggregate sequence representation for classification tasks. Sentence pairs are packed together into a single sequence. We differentiate the sentences in two ways. First, we separate them with a special token ($[\text{SEP}]$). Second, we add a learned embedding to every token indicating whether it belongs to sentence A or sentence B. As shown in Figure 1, we denote input embedding as E , the final hidden vector of the special $[\text{CLS}]$ token as $C \in \mathbb{R}^H$, and the final hidden vector for the i^{th} input token as $T_i \in \mathbb{R}^H$.

11 tokens [CLS+10]
10 word two sentences

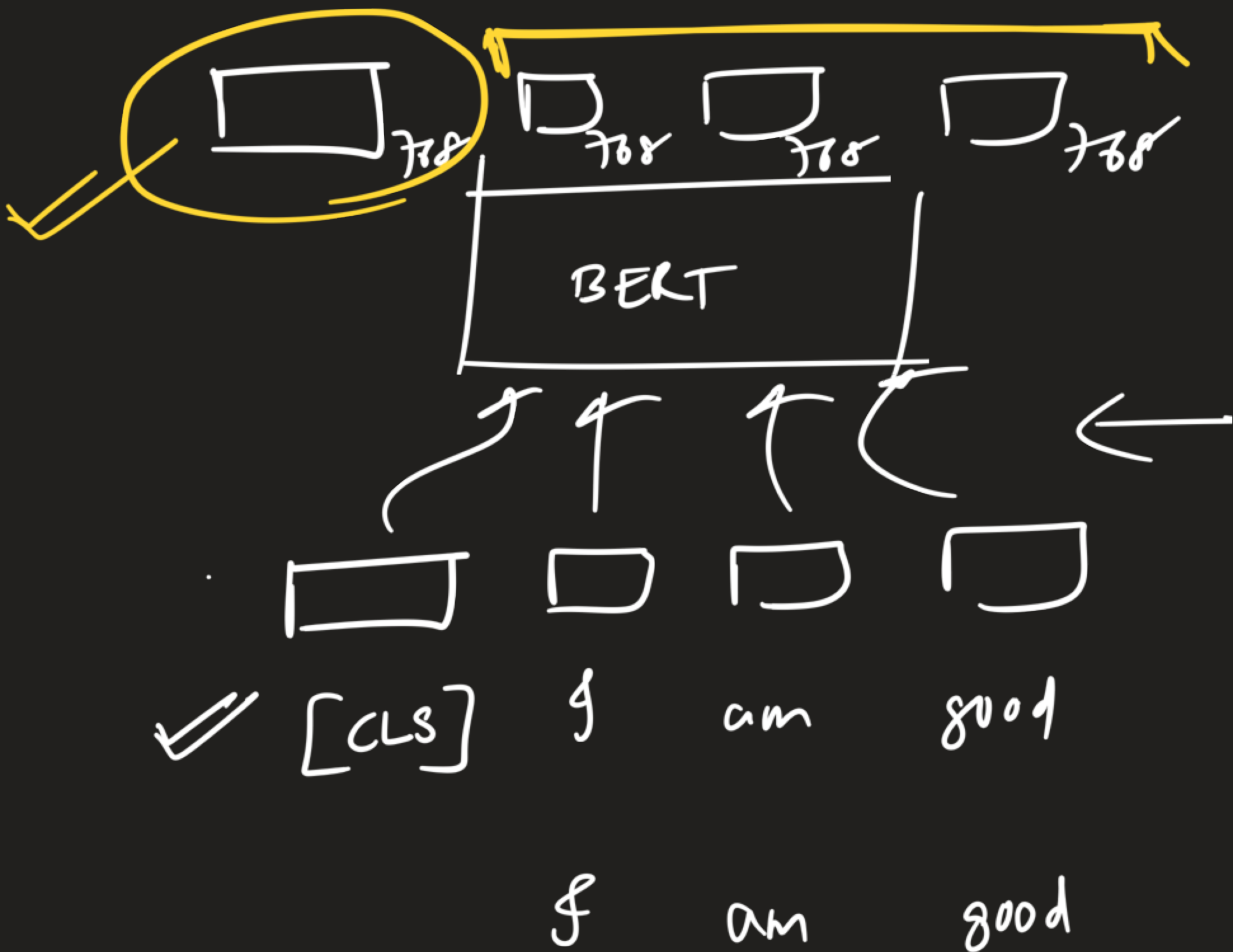
tokens

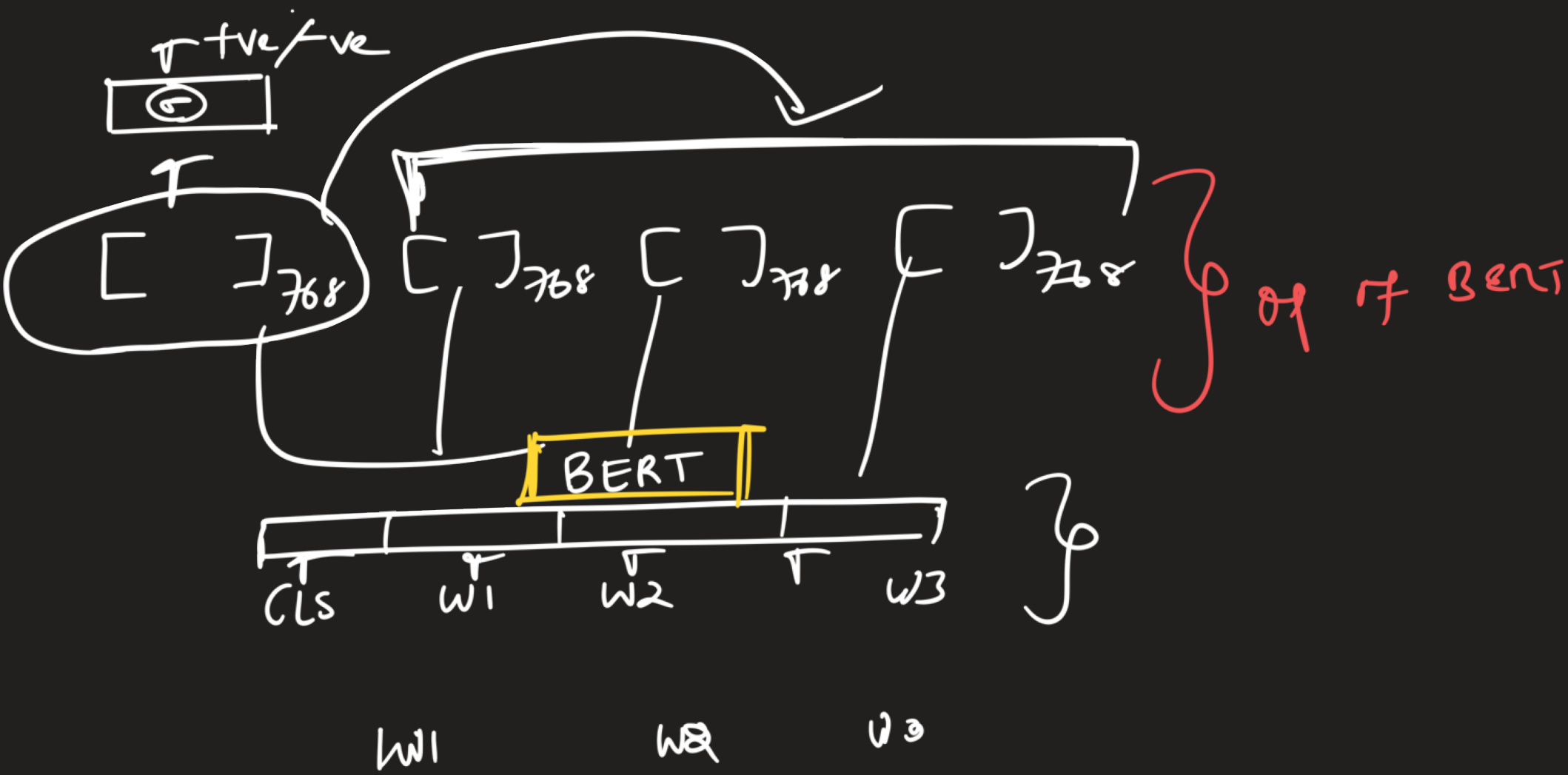


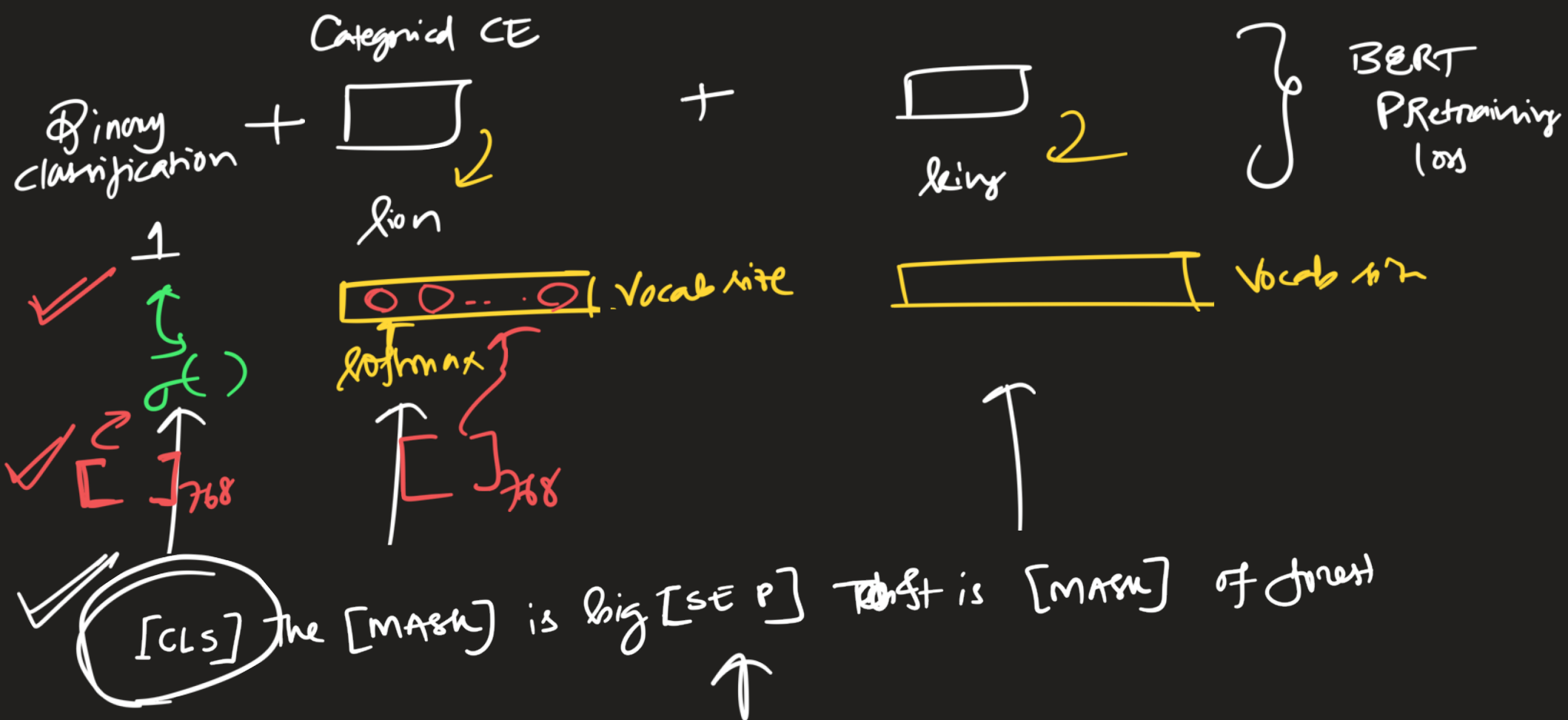
Pre-training

11×768

11×768







S — Arijit is a ^{MASK} good singer
↓
good

x	y
Arijit is a MASK singer	good

huge chunk of Internet

19:21

Self-Supervised

Pre-training objectives (loss)

1] Masked Language Model obj

2] Next Sentence Prediction obj

capture relationship betⁿ two sentences

Q] What is capital of India? New Delhi
S1 S2

Cat is a mammal

Lizard is a reptile

1] MLM/Cloze

Words within a sentence give sense to the sentence

S: input: Arijt is a good singer
op: good

[MASK]
[MASK]
BERT
↓
good

Training data
generation for
MLM

Masked LM and the Masking Procedure Assuming the unlabeled sentence is my dog is hairy, and during the random masking procedure we chose the 4-th token (which corresponding to hairy), our masking procedure can be further illustrated by

- 80% of the time: Replace the word with the [MASK] token, e.g., my dog is hairy → my dog is [MASK] — Randomly pick a word and mask
- 10% of the time: Replace the word with a random word, e.g., my dog is hairy → my dog is apple
- 10% of the time: Keep the word unchanged, e.g., my dog is hairy → my dog is hairy. The purpose of this is to bias the representation towards the actual observed word.

*Semi supervised learning

X	Y
my dog is [mask]	hairy

The cricketer is good. The owl sleep is day.

$\mathcal{O}(\text{NoNext})$ cricketer

owl

↑

↑

↑

[CLS] The [MASK] is good [SEP] The [mask] sleeps in day

The lion is big. It is king of forest

1 (isNext)

lion

king

↑

↑

↑

⊗ [CLS] The [mask] is big. It is [mask] of forest

S1 . S2

NSP
↓
BC ✓
MC
Regression

logloss

Next Sentence Prediction The next sentence prediction task can be illustrated in the following examples.

Input = [CLS] the man went to [MASK] store [SEP] S1
 he bought a gallon [MASK] milk [SEP]
Label = IsNext ✓ 1

Input = [CLS] the man [MASK] to the store [SEP] S1
 penguin [MASK] are flight ##less birds [SEP]
Label = NotNext S2

↑
0

The tiger is big
It's the king of
jungle

↓

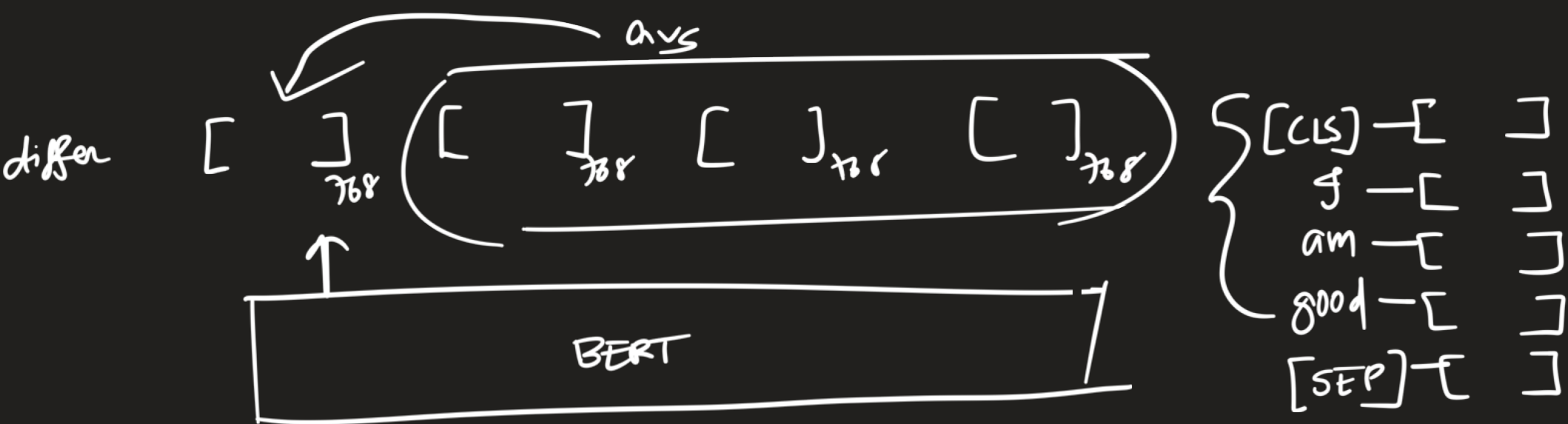
~~king~~

The [mask] is big
It is [man] of
jungle ~~king~~
isNext ✓

The lion is in Sahara

F1 score



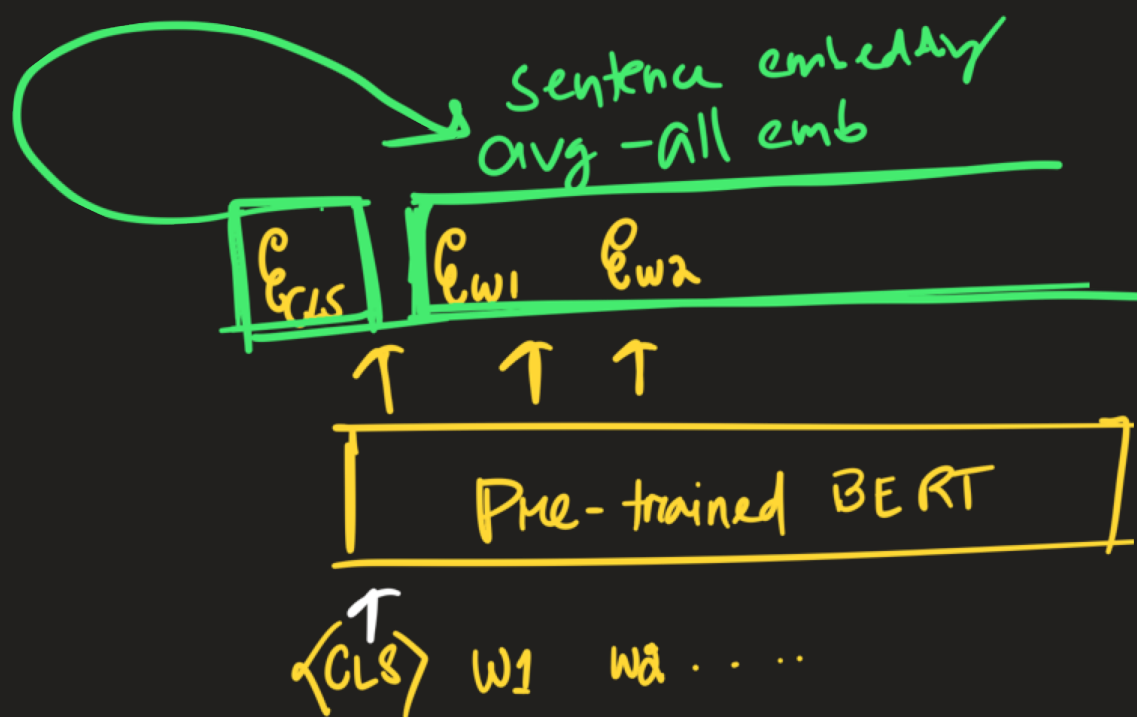


Same { $[0 \dots 0]$ $[0 \quad 0]$ $[0 \quad 0]$ $[0 \quad 0]$ }

CLS g am good

tv/-ve ✓

The song is good $\rightarrow$ +ve/-ve



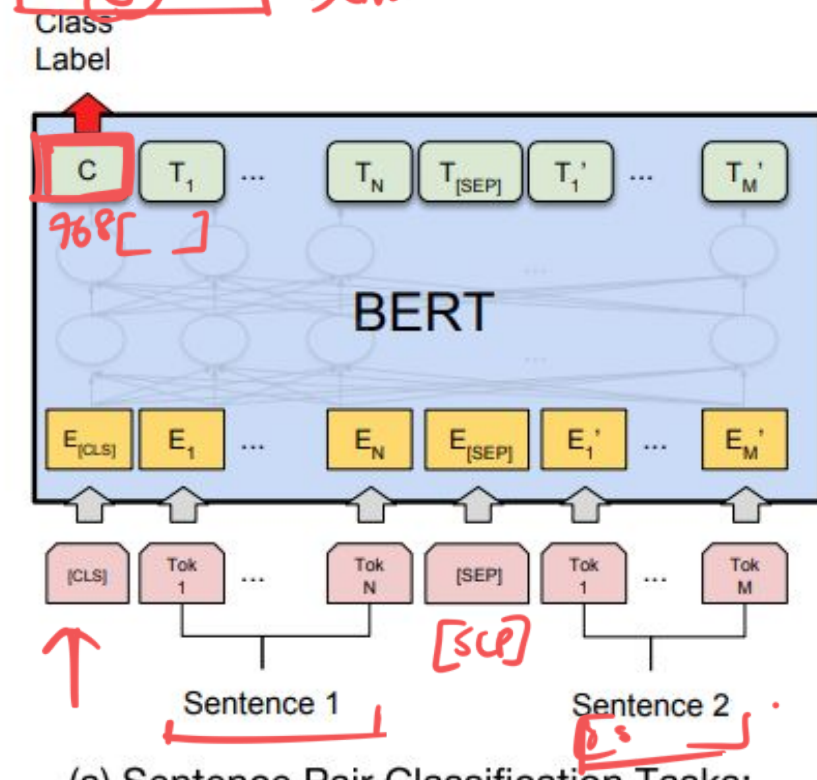
Sentence embedding

Fine Tuning-1

Log loss

+ve -ve

0 Dene



(a) Sentence Pair Classification Tasks:

Sentence Pair

X

S1 S2

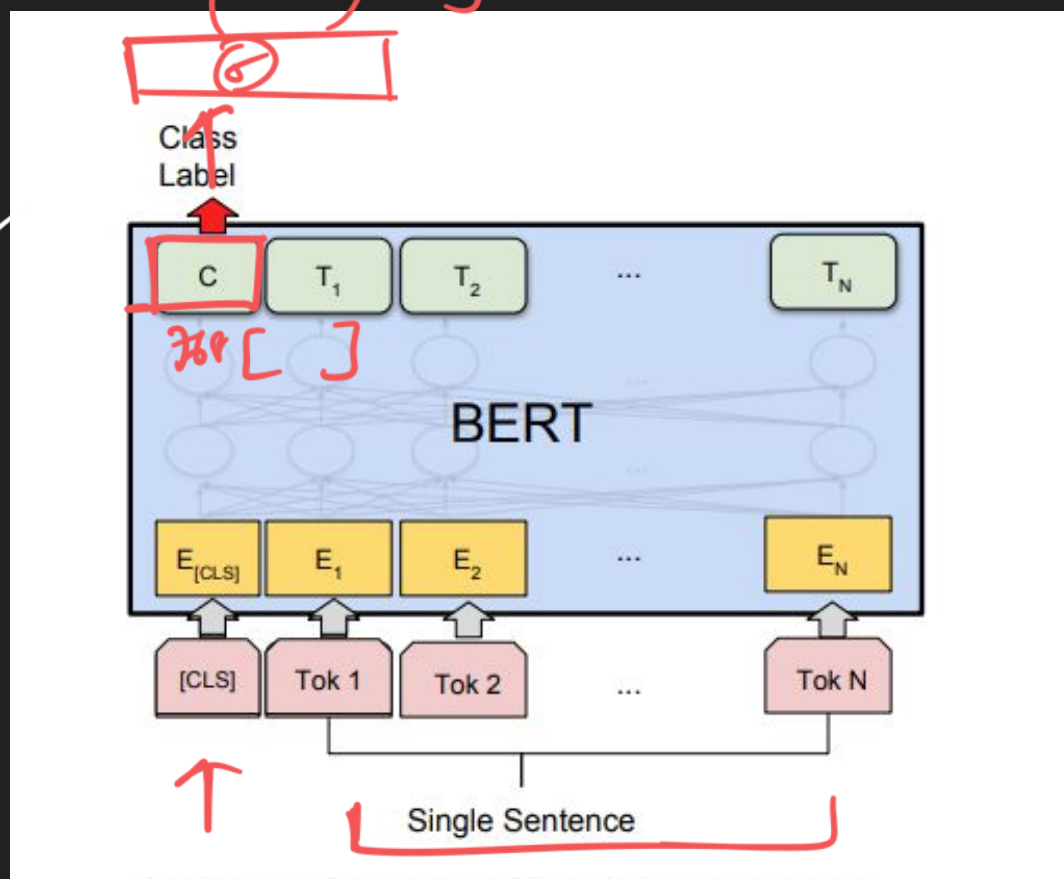
S1 S2

+ve
from same
pair

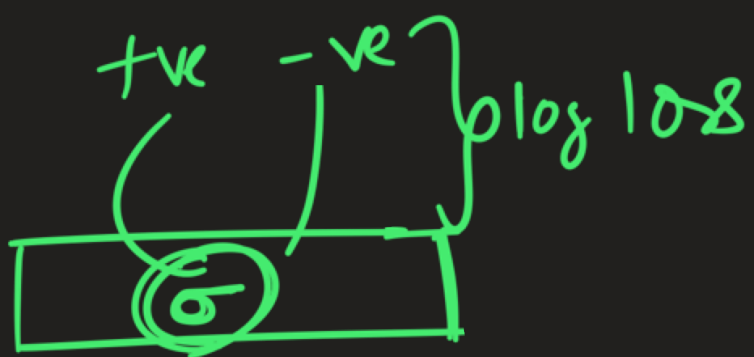
-ve
not from
same pair

Fine Tuning-2 (Sequence classification)
 $\{ \text{true} \rightarrow \text{true} \} \text{ loss}$

X	Y
Document	
Virat Kohli	Sports
Modi	Politics
	Religion



X	Y
Arijt sings sad	+
Satya sings bad	-



σ
 $[CLS]$ $\uparrow$

σ	σ	σ	σ
σ_{Arjit}	σ_{sing}	σ_{good}	
$\uparrow$	$\uparrow$	$\uparrow$	

BERT

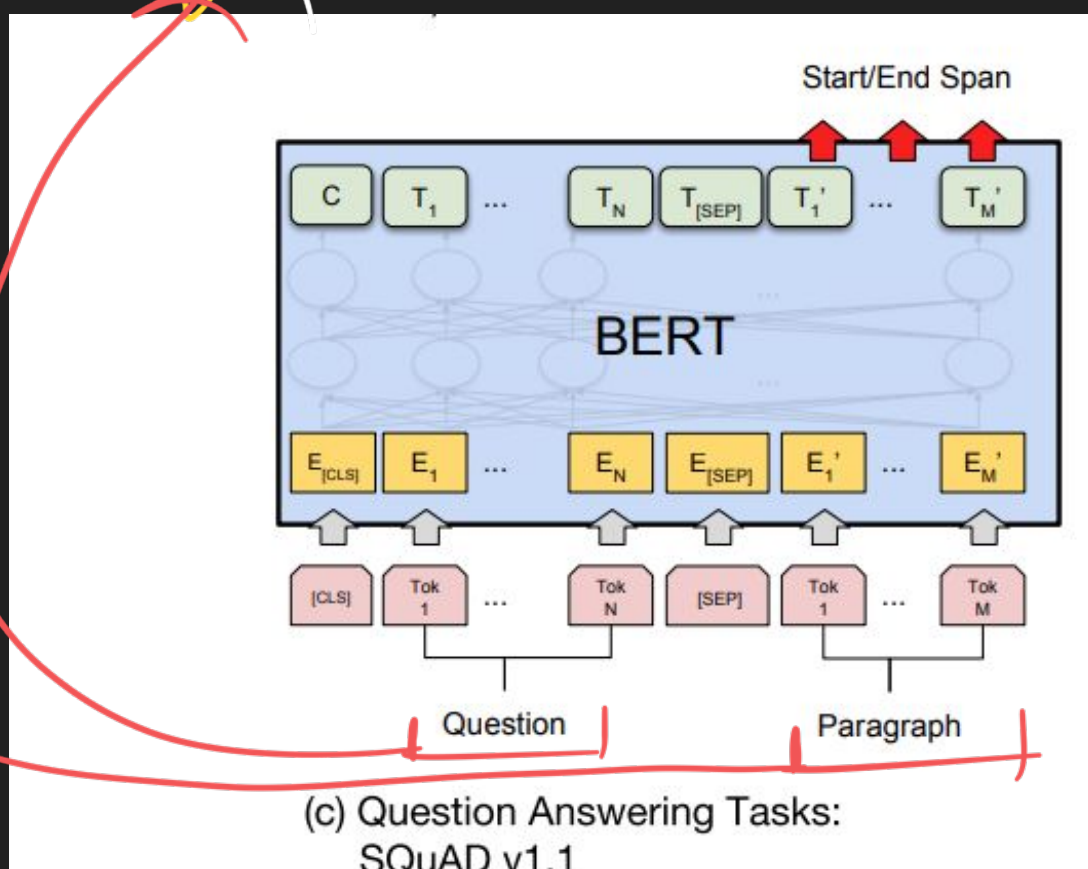
$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$
$[CLS]$	Arjit	sing	good

Sequence classification

Fine Tuning-3

Paragraph - Delhi is polluted. Delhi is known for CR.
Delhi is capital of 7th largest country.
India is 7th largest country of world

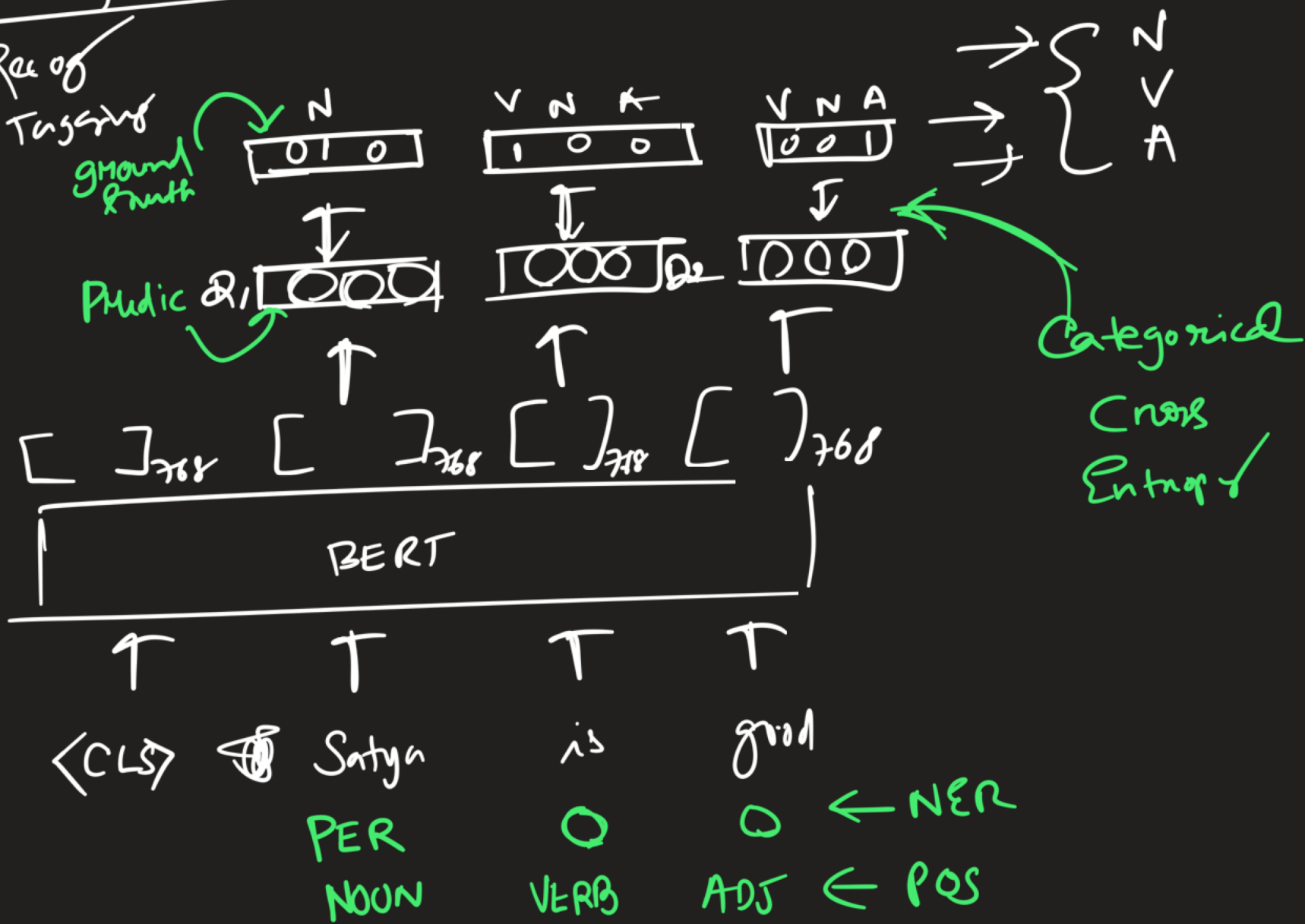
Question - What is capital of India



Disclaimer - not popularity comp

Token level classification

Named Entity Recog
Part of Speech Tagging



512 words as input

NER

ORL
↑
Microsoft + microsoft

cased

Great
great

heavy

uncased

great

sentiment

great } ✓
Great } +ve

8
am
word

19
20
100

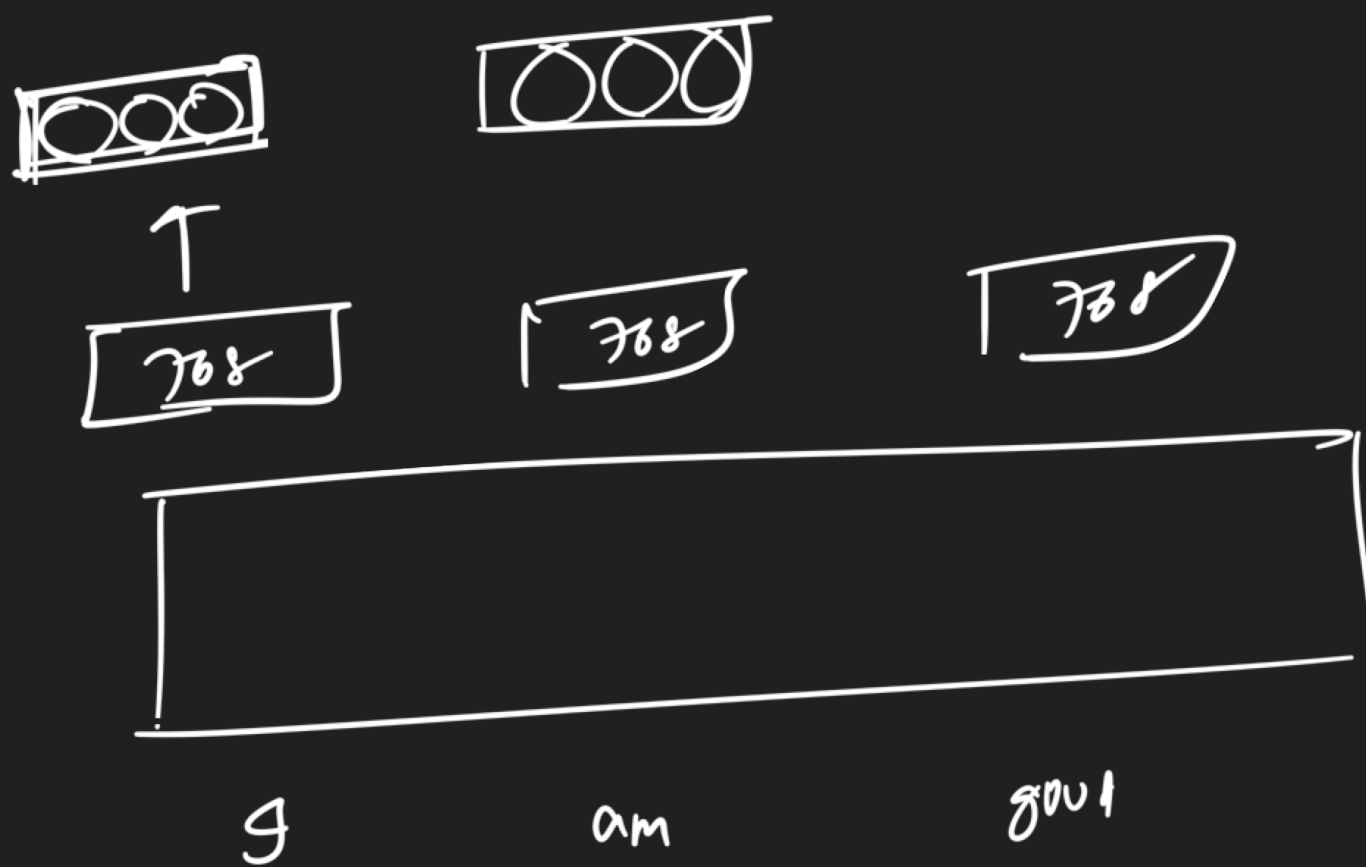
{ 19 - []
20 - []
100 [] }

1
CLS 500

" } 512
PAD

Token Type ✓✓
SD

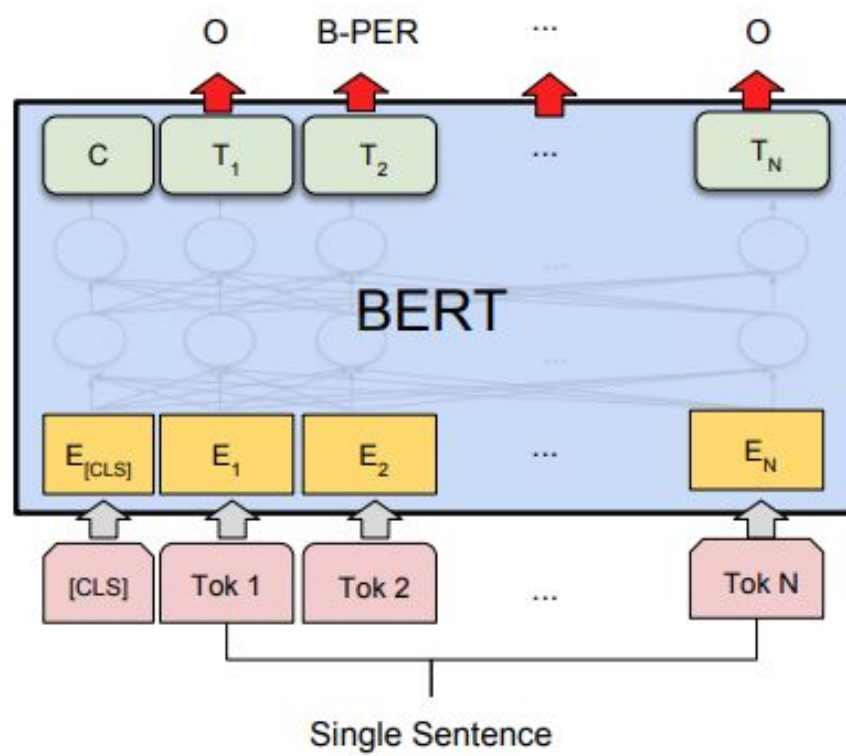
Token type D O O 1 1 1 } demarcate sentence
W1 wa W3 W4 W5 W6
S1 S2



0 — other
1 — Category
2 — PAD

Token Classification Head

Fine Tuning-4



(d) Single Sentence Tagging Tasks:
CoNLL-2003 NER

TOKEN CLASSIFICATION

Tokens: ["The", "intersex", "differences", "in", "colorectal", "cancer", "risks", "have", "implications"]

Token ids: [1996, 6970, 3366, 2595, 5966, 1999, 3609, 22471, 2389, 4456, 10831, 2031, 13494]

Label entity: "colorectal cancer"

Train label: [0, 0, 0, 0, B, B, 0, 0, 0]

Label: [0, 0, 0, 0, 1, 1, 0, 0, 0]

	BERT	RoBERT	DistilBERT	XLNet
Size (millions)	Base: 110 Large: 340	Base: 110 Large: 340	Base: 66	Base: ~110 Large: ~340
Training Time	Base: 8 x V100 x 12 days* Large: 64 TPU Chips x 4 days (or 280 x V100 x 1 days*)	Large: 1024 x V100 x 1 day; 4-5 times more than BERT.	Base: 8 x V100 x 3.5 days; 4 times less than BERT.	Large: 512 TPU Chips x 2.5 days; 5 times more than BERT.
Performance	Outperforms state-of-the-art in Oct 2018	2-20% improvement over BERT	5% degradation from BERT	2-15% improvement over BERT
Data	16 GB BERT data (Books Corpus + Wikipedia). 3.3 Billion words.	160 GB (16 GB BERT data + 144 GB additional)	16 GB BERT data. 3.3 Billion words.	Base: 16 GB BERT data Large: 113 GB (16 GB BERT data + 97 GB additional). 33 Billion words.
Method	BERT (Bidirectional Transformer with MLM and NSP)	BERT without NSP**	BERT Distillation	Bidirectional Transformer with Permutation based modeling

✓
Rope

12000
millions }



Arijit sings in Hindi. He is good. ↓

} S12

Token type
Tns

0 0 0 0 0 0 0

Token type Tns

India is rich; Apollo is spacecraft
S1 S2
0 0 1 1 1